



Local fiscal pressure and new firm entry

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ARTICLE INFO

Keywords:

Local fiscal pressure
New firm entry
Governmental intervention
Credit availability

ABSTRACT

Local governments typically implement various policies to attract new enterprises to their jurisdictions. However, under fiscal constraints, some of their behavioral changes may impact the willingness of new businesses to establish themselves locally and could even lead to counterproductive outcomes. This study, utilizing data from prefecture-level cities in China, identifies a negative correlation between local government fiscal pressure and the entry of new enterprises. This conclusion withstands a series of robustness tests. After eliminating potential competing mechanisms, it is found that during periods of fiscal difficulty, local governments channel more financial resources towards large local enterprises to achieve their objectives. Such biased guidance diminishes credit availability for small and medium-sized enterprises, consequently inhibiting the formation of new enterprises. Heterogeneity analysis reveals that this suppressive effect is less pronounced in regions with higher levels of inclusive financial development or lower degrees of government intervention, and when considering only the entry of non-state-owned enterprises. Spatial econometric analysis demonstrates an “expulsion effect,” where regions with higher fiscal pressure drive newly established enterprises to areas with lower fiscal pressure. Our research provides empirical evidence for the theory of the governmental “interventionist hand” and reveals the corresponding consequences. It thereby offers insights and reflections for governments on how to effectively facilitate local entrepreneurship, as well as ensuring proactive and viable policies amid fiscal constraints.

1. Introduction

Entrepreneurial activity, represented by the creation of new companies, is one of the driving forces behind economic growth (Bertrand & Kramarz, 2002; Gourio et al., 2016; Reynolds, 1999). On the one hand, newly entrant enterprises bring about a “creative destruction” effect (Schumpeter, 1942) by introducing innovative technologies or novel forms of industrial organization, causing continuous disruption to the existing inefficient market equilibrium. As more entrepreneurs enter the market, economic rent decreases, and the market tends towards a more efficient equilibrium. On the other hand, the influx of new companies fosters competition, compelling incumbents to enhance their levels of innovation and productivity (Aghion et al., 2004, 2009). This, in turn, contributes to productivity improvements in an aggregated level (Foster et al., 2008). Given this perspective, the entry of new businesses within a region to some extent mirrors the local economic vitality.

Facilitating the entry of new firms is typically one of the major objectives for many governments, especially at the subnational level. This endeavor is often pursued through methods such as reducing entry costs

(Alesina et al., 2005), implementing tax cuts (Venâncio et al., 2020), providing subsidies (Accetturo, 2022), and more. Empirical findings also corroborate that the entry of newly established enterprises contributes positively to regional economic progression, exemplified by an increase in employment (Kirchhoff & Phillips, 1988; Fritsch & Mueller, 2004; Van Stel & Suddle, 2008), as well as by fostering advancements in overall productivity (Audretsch & Keilbach, 2004; Erken et al., 2018) or innovation (Modrego et al., 2015; Gumbau Albert, 2017). However, a substantial body of literature uncovers that in the presence of fiscal constraints, government behavior tends to be somewhat distorted. For example, in response to fiscal pressure, the government may expand informal charges (Liu, 2018), enhance tax enforcement (Chen, 2017), transfer expenditure obligations to companies (Wen et al., 2023), or intervene in the allocation of financial resources within its jurisdiction (Sun & Wang, 2020; Zhu et al., 2020). Consequently, do these behavioral changes of the government under fiscal pressure reduce the attractiveness for the entry of new enterprises, or even yield a repelling effect? What is the underlying mechanism at play here? The current body of literature fails to respond to these questions. Furthermore, the

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<https://doi.org/10.1016/j.irfa.2025.103937>

Received 18 July 2024; Received in revised form 16 January 2025; Accepted 16 January 2025

Available online 19 January 2025

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exploration of such issues holds practical significance for the government in inspiring entrepreneurship and mitigating the negative impacts of fiscal pressures.

China's development experience may provide preliminary insights into the correlation between entrepreneurship and economic growth. From the late 1980s when the legitimacy of private sector was first acknowledged, to the end of 2021 when private enterprises constituted 92.1 % of all enterprises nationwide, China's economy underwent a rapid expansion, ultimately propelling it to the position of the world's second-largest economy. For an extended period, governments at various levels in China have maintained a proactive stance towards encouraging entrepreneurship. For instance, following Premier Li Keqiang's introduction of the concept of "mass entrepreneurship and innovation" in the 2015 Government Work Report, the State Council has issued a series of policy directives aimed at guiding entrepreneurial endeavors throughout the entire society. At the subnational level, prefectural-level city governments have implemented numerous preferential policies tailored for newly registered business, such as offering low-interest loans, startup subsidies, rent subsidies for venues, and social security subsidies for entrepreneurs.¹

However, the fiscal condition of local governments in China has deteriorated sharply in recent years. Under the framework of fiscal decentralization,² local governments in China have long grappled with the challenge of aligning tax revenues with diverse expenditure needs. Over the past decade, this challenge has been intensified by inclusive policies introduced by the central government and the effects of public health events. Notably, the implementation of the comprehensive "Business Tax to Value-Added Tax" reform in 2016, followed by a series of "tax cuts and fee reductions" policies in 2018, along with the outbreak of the COVID-19 pandemic at the end of 2019, have all contributed to a significant reduction in the fiscal revenue of local governments (Guo & Shi, 2021; Lin et al., 2024; Wang et al., 2022). In addition, the 2014 budget law imposed strict limits on the annual debt issuance quotas for local governments. That same year, the State Council's issuance of "Document No. 43" further reinforced these financial constraints by severing the ties between local governments and their financing vehicles. These measures collectively imposed significant constraints on the avenues available to local governments for financing through debt. However, daily government operations and essential public expenditures exhibit a high degree of rigidity, often referred to as a "ratchet effect" (Hercowitz and Strawczynski, 2004; Lee & Plummer, 2007). As a result, the financial strain on local governments continues to escalate annually, as illustrated in Fig. 1. Therefore, China offers an ideal scenario for analyzing the correlation between government fiscal pressure and the entry of new firms.

In the figure, the left axis represents the values of fiscal revenue and expenditure, measured in units of 100 million yuan. The right axis corresponds to the fiscal self-sufficiency rate, which is calculated as fiscal revenue divided by fiscal expenditure. From the figure, we can observe that the gap between fiscal revenue and expenditure for China's local governments is expanding year by year, and the fiscal self-sufficiency rate shows a clear declining trend, both of which indicate a growing fiscal pressure.

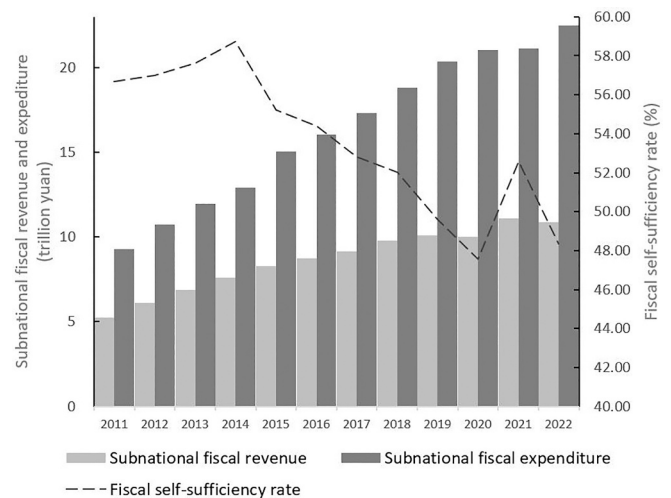


Fig. 1. Trends in China's local government fiscal revenue and expenditure over the years.

Using economic indicator data for China's prefecture-level cities spanning the years 2011 to 2021, our analysis reveals that local government fiscal pressure significantly hinders the entry of new enterprises. This finding maintains its robustness when incorporating alternative metrics for both explanatory and dependent variables, accommodating time lags in the explanatory and control variables, and redefining the geographical placement of firms. In the heterogeneity analysis, this inhibitory effect is consistently observed in the influx of non-state-owned enterprises (non-SOEs), while it is not evident in the context of state-owned enterprise (SOE) entries. Moreover, the primary inference stands firm in city samples characterized by intense financial resource manipulation or nascent inclusive financial development. During our investigation into the underlying mechanisms, we ruled out the potential of local fiscal pressures influencing new enterprise entries via government subsidies, tax enforcement or the local business climate. Instead, following a comprehensive examination of various databases related to local enterprises, our analysis suggests a "credit availability differentiation" pathway, wherein fiscal stress affects the scale of bank credit accessible to local businesses of varying sizes.

To alleviate the potential endogeneity concerns in the main regressions, we introduce a causal identification framework based on a natural experiment. The implementation of the Fifty-Fifty Revenue Sharing Reform for Value-Added Tax (VAT) (hereafter VAT revenue sharing reform) in May 2016 significantly reduced the fiscal revenue of local governments in China, particularly impacting those cities where business tax constituted a significant proportion of fiscal revenue. This policy was part of a broader suite of tax reforms pursued by the Chinese central government, largely independent of the prevailing economic conditions and without directly affecting enterprise inflows or outflows. Therefore, it can be considered an exogenous shock that intensified fiscal pressures on local governments, allowing us to delineate control and treatment groups based on the varying intensities of policy impact. Subsequently, we employ a Difference-in-Differences (DID) model to validate the effect of fiscal pressure on enterprise entry. Furthermore, to explore the spillover effects of local fiscal pressures on enterprise entry in neighboring regions, we use spatial econometric models for analysis, taking into account the potential spatial interdependence between new business registrations and fiscal pressures across cities. After conducting relevant tests, the General nesting spatial model (GNSM) was selected. Besides, to ensure a more nuanced and accurate representation of these spatial relationships, our approach incorporates an array of diverse spatial weight matrices. The empirical results affirm the conclusion drawn from the main regression and further unveil an "expulsion effect" of fiscal pressure in neighboring cities on the establishment of new

¹ For example, Shenzhen municipal government provides one-time subsidies ranging from 10,000 to 100,000 RMB for startup companies, along with a monthly social insurance subsidy of 840 RMB and an approximate 1000 RMB housing subsidy for entrepreneurs.

² In 1994, with the aim of encouraging local governments to drive economic growth and increasing the central government's revenue (Sima et al., 2023), China implemented the tax sharing system. This system clearly defined the proportions of tax revenue allocation between the central and local governments, as well as their respective expenditure responsibilities. Consequently, local governments were largely required to manage their finances independently, bearing both the profits and losses.

enterprises.

Overall, this study, by examining the correlation between local government fiscal pressure and the entry of new firms, not only offers novel insights into the widely debated academic topic of factors influencing firm entry (Bruhn, 2011; Dai & Si, 2018; Kong & Qin, 2021; Lee & Mukoyama, 2015), but also complements the literature on the impacts caused by local fiscal pressures (Han & Kung, 2015; Jiang et al., 2022; Xi et al., 2017; Xiong et al., 2024). Furthermore, this paper sufficiently considers and addresses the challenging issues of endogeneity and spatial autocorrelation in the study of local fiscal pressure impacts. Finally, our study provides empirical evidence relevant to the discussion on government intervention in financial resources (Cole, 2009; Dinç, 2005; Englmaier & Stowasser, 2017; Zhu et al., 2020), and further examines the consequences of such biased governmental actions at the macroeconomic level.

The rest of this paper is organized as follows. Section 2 reviews relevant literature and proposes research hypotheses based on the logical framework established in existing studies. Section 3 presents the data, variables and regression models employed in this study. Section 4 details and analyzes the primary empirical findings. Section 5 conducts extended research. Section 6 provides the conclusion of the paper.

2. Literature review and hypothesis development

2.1. Firm entry

The concept of “new firm entry” discussed in this paper refers to the process by which newly established enterprises register and commence operations in a specific region. Alternatively, it can be interpreted as an increase in the number of new enterprises in a given area, reflecting the entrepreneurial vitality of that region. Studies concerning firm entry primarily centers on the factors that affect the initiation of a business in a specific place. This encompasses factors like personal traits of entrepreneurs, macroeconomic indicators, and government policies. Firstly, the initial founding of a business is greatly dependent on entrepreneur’s personal characteristics, where aspects like their gender (Klapper & Parker, 2011; Yang & Aldrich, 2014; Thébaud, 2015; Minniti, 2009), age (Evans & Leighton, 1989; Karoly & Zissimopoulos, 2004; Kautonen et al., 2014; Levesque & Minniti, 2006), educational level (Oosterbeek et al., 2010; Vanevenhoven & Liguori, 2013), and financial status (Bueria, 2009; Gujrati et al., 2019) play pivotal roles. Secondly, alterations in the macroeconomic environment will also influence the entry of businesses into the region. For instance, empirical studies demonstrate that factors such as the growth of the digital economy (Zhao et al., 2022), diminishing trade policy uncertainty (Cui & Li, 2023), enhanced governmental efficiency (Du et al., 2020), and the adoption of industrial robots (Chen et al., 2023) are all conducive to the entry of new enterprises in a region. Conversely, escalating population aging (Feng & Li, 2023) and rising political uncertainty (Chen et al., 2020) have generally been observed to exert a negative effect. Thirdly, certain specific policies enacted by the government, though not explicitly aimed at the entry of new companies, will nonetheless exert an indirect impact. Da Rin et al. (2011) affirmed that variations in the tax rates imposed on companies are closely associated with corporate entry. Li (2020) posited that the reform allowing local governments in China to transfer land use rights through market transactions has increased the likelihood of enterprise entry by enhancing urban land use efficiency, promoting the development of urban infrastructure, and reducing urban rent-seeking corruption. Kong and Qin (2021) identified that China’s anti-corruption campaign has improved the entrepreneurial environment, primarily by curtailing rent-seeking opportunities and elevating the quality of government administration. Moreover, Li et al. (2022) discovered that China’s initial round of environmental inspections conducted by China’s central government notably decreased the entry of companies into high-polluting industries. Besides, Bai et al. (2022) confirmed the impact of the national innovative city pilot policy in fostering innovation, most

prominently evidenced by the surge in entrepreneurial activity within the pilot cities.

2.2. The impact of local fiscal pressure

This study is also relevant to the literature on the impact of local fiscal pressure on regional macroeconomic environments or corporate behavior. When confronted with substantial fiscal deficits or a large scale of debt, the government’s foremost priority lies in devising strategies to augment revenue or cutting expenditures (Miller & Russek, 1997; Nelson & Balu, 2014). At the same time, it must contemplate methods to invigorate local economic development (Eisinger, 1988) and achieve political accomplishments that support official promotion (Que et al., 2019), all executed within the constraints of limited financial resources. Consequently, specific government actions motivated by self-interest have the potential to impact the regional macroeconomic and ecological environment, as well as the behavior of micro-enterprises within their jurisdictions. In their study utilizing data from China, Han and Kung (2015) observed that when local governments experience a decline in fiscal revenue, they are more inclined to foster the growth of local real estate and infrastructure construction sector, as opposed to focusing on the industrial development. Slightly differing in their view, Xi et al. (2017) claimed that local governments, in response to financial pressure arising from a reduction in the VAT share ratio, will intensify their support for overcapacity industrial enterprises that can generate substantial VAT revenue. Ruan and Zhao (2022) uncovered that in response to fiscal pressure, governments in low-income regions tend to intensify tax collection and maintain a reliance on agricultural production. In contrast, governments in high-income regions persist in attracting enterprises through tax incentives and introduce substantial capital investments. These distinct tactics lead to regional disparities in economic growth. Furthermore, when constrained by limited fiscal options, such as tax incentives or subsidies, due to fiscal pressures, the government may resort to alternative resources, such as land resources and local financial resources, to attain its objectives. Qi et al. (2020) revealed that under financial pressure, China’s local governments tend to transfer land use rights at low prices as a strategy to attract investments. Zhu et al. (2020) contended that when facing fiscal pressures, local governments tend to interfere in the distribution of local credit resources and direct a larger share towards the state-owned economic sector. Beyond their economic effects, fiscal pressures also give rise to environmental challenges. Escalating fiscal pressures often cause a decrease in governmental attention to environmental matters, a rise in energy usage, and diminished advancements in green technology, subsequently heightening carbon emission levels (Xue et al., 2023). The research by Bai et al. (2019) indicates that regional tax competition, driven by local government fiscal pressures, negatively affects the local environment and reduces environmental quality in neighboring areas.

In terms of the impact of fiscal pressure on corporate behavior, Cai et al. (2022) asserted that a reduction in government fiscal pressure results in a decreased tax burden for businesses, thereby improving their profitability and performance. According to the findings of Wen et al. (2023), the fiscal pressure on local governments restricts corporate earnings management. Under fiscal stress, local governments, on the one hand, intensify tax enforcement, deterring businesses from reducing their reported earnings; while on the other hand, they transfer expenditure responsibilities to corporations, diminishing firms’ inclination to artificially increase their earnings. Peng et al. (2023) argued that a range of government actions in response to fiscal stress result in elevated financing costs for businesses, diminished internal cash reserves, and foster a tendency towards corporate financialization. As a consequence, all of these changes contribute to a decrease in corporate risk-taking. In addition, financial pressure may also affect a company’s environmental performance. Researchers hold that when confronted with fiscal strain, local governments tend to shift their priorities, such as relaxing environmental regulation (Zhou et al., 2023) and escalating inter-regional

tax competition (Bai et al., 2019). These shifts in focus consequently encourage businesses to prioritize production over environmental protection. Moreover, the financial pressures faced by local governments often result in increased non-tax burdens or reduced subsidies for environmental protection. Consequently, it increases companies' financial constraints, potentially reducing their capacity and inclination to invest in green technological innovations (Kong & Zhu, 2022) or environmentally friendly investments (Liu & Ren, 2023). Jiang et al. (2022) further noted that in fiscal pressure, local governments can distort energy prices to boost industrial development, thus decreasing enterprises' energy utilization efficiency.

2.3. Hypothesis proposal

The influence of government actions on micro-enterprise decision-making is a frequently debated subject within academic discourse. Such insights hold practical implications for the establishment of a "proactive government." As demonstrated in Section 2.2, numerous scholars have engaged in discussions on this topic by examining changes in government behavior resulting from fiscal pressures. Regrettably, to the best of our knowledge, there exists a dearth of scholarly research exploring the influence of fiscal pressure faced by local governments on firm entry into the locality. However, existing empirical findings suggest that governments, when confronted with fiscal difficulties, may adopt measures such as reducing subsidies (Ilyas et al., 2022; Phomsoda et al., 2021), enhancing tax enforcement (Chen, 2017; Ruan & Zhao, 2022), and intervening in the allocation of financial resources (Sun & Wang, 2020; Zhu et al., 2020). These actions are likely to impede the creation of firms in the region. Considering this context, we present the primary hypothesis, H1, of our study.

H1. Fiscal pressure on local governments impedes the entry of new enterprises.

Research in the literature establishes that governments may manipulate the distribution of financial resources within their jurisdictions to serve their own objectives. Such actions may stem from the inherent responsibilities of governments, including efforts to stimulate economic growth (Andrianova et al., 2012; Gerschenkron, 1962) and rectify market failures (Besley, 1994; Spratt, 2009; Stiglitz, 1989). Alternatively, they might also be driven by self-serving motives, such as seeking to gain more votes (Cole, 2009; Dinc, 2005; Micco et al., 2007) or to facilitate the promotion of officials (Ji et al., 2014; Wang et al., 2019). In the specific context of China, although the central government possesses the access and supervisory rights over financial institutions, local governments still retain the capability to intervene in the credit allocation decisions of local financial institutions, either through holding stakes (Xu, 2018) or by employing administrative means (Yao & Peng, 2013). Zhou (2003) highlighted that local financial resources have evolved into what can be termed "secondary fiscal reserves" for local governments in China. Many scholars have also confirmed the phenomenon that local governments in China may mobilize local financial resources for purposes of political advancement or inter-regional competition (Qian et al., 2011; Sun & Wang, 2020; Tian et al., 2023).

In addition, existing literature indicates that local governments in China are pro-business (Li & Zhou, 2005), particularly towards major and key enterprises within their jurisdictions (Nie et al., 2022; Wang et al., 2014). The main logic behind this comprises three key points. Firstly, these entities play a pivotal role in local economy by not only markedly propelling regional economic expansion and facilitating substantial investments but also serving as the principal contributors to local tax revenues. Secondly, large-scale enterprises often serve as the core of local industrial chains (Grzybowska, 2011; Yan, 2013), which are capable of attracting industrial clusters (Alfaro et al., 2019). Through mechanisms such as supply chain integration, knowledge spillovers, and the creation of labor pools, they exert a positive influence on the development of local small and medium-sized enterprises (SMEs)

(Jannati, 2020; Lu et al., 2023; Lu et al., 2024; Owen-Smith & Powell, 2004; Ye & Zhuang, 2022). Thirdly, large local enterprises usually maintain close ties with local governments, sometimes even involving direct equity relationships. Consequently, they actively lobby for their own corporate interests and seek to influence the allocation of resources in their favor (Ji & Zhang, 2024; Luo & Tang, 2009).

In light of this, under fiscal pressures, the inclination of local governments to favor large enterprises in resource allocation may be intensified. In scenarios where fiscal resources are scarce, local governments find it challenging to rely solely on public funds and fiscal tools to stimulate investment and promote local economic development (Di Liddo et al., 2019; Tang et al., 2023). At this juncture, local governments may offer greater conveniences to local large-scale enterprises, with the aim of enhancing their role in driving the local economy. Moreover, when confronted with fiscal constraints, local governments are further incentivized to engage in tax base protection (Chung & Williams, 2021; Ruan & Zhao, 2022). As a result, they are inclined to allocate additional resources to key revenue-generating enterprises to maintain their normal and sustained operations. Lastly, in times of fiscal distress, local governments may postpone or even default on payments owed to contractors engaged in major infrastructure projects, which are typically local large enterprises (Li et al., 2023). As a form of compensation, to some extent, the government may also channel a greater allocation of credit resources towards these enterprises through the exercise of administrative power.

The preceding analysis raises an important question: under fiscal pressure, is it likely that local governments' motivation to intervene in financial resources will increase, resulting in a larger allocation to local large enterprises? If this is the case, local SMEs' credit resources will most likely be crowded out. SMEs in China have traditionally faced financial repression due to a lack of collateral assets and the absence of a well-developed multi-tier capital market (Li et al., 2024; Lu et al., 2022). When their potential credit space is further squeezed, the accessibility to financial resources becomes even more restricted. Given that financial resources play a crucial role in firm establishment and entrepreneurial activities (Di Patti & Dell'Arciccia, 2004), increased difficulty in accessing such resources, along with the resulting concerns over future credit availability, will directly contribute to a decline in the number of new local business registrations. Based on this reasoning, we put forth the second hypothesis (H2).

H2. Local fiscal pressure leads to the crowding out of credit resources available to local SMEs.

Expanding revenue sources and reducing expenditures are almost reflexive strategies for governments when facing fiscal pressures (Nelson & Balu, 2014; Wolman, 1983). As a form of direct government expenditure, subsidies dispensed by the government are prone to diminution during intervals of fiscal adversity (Ilyas et al., 2022; Phomsoda et al., 2021). In fact, the reduction of subsidies can, indeed, sometimes result directly in significant expenditure savings for the government. For instance, in 2015, Indonesia's reform of diesel and gasoline subsidies led to a notable saving of approximately 15.6 billion US dollars, marking a significant reduction in government expenditure for that year. Similarly, Egypt undertook a substantial cut in its fossil fuel subsidies in 2014, alleviating a considerable fiscal strain that had previously consumed more than 20 % of its 2013 budget. Nevertheless, given that government subsidies play a pivotal role in the sustainability of startup companies (Duhautois et al., 2015), the reduction of these subsidies for local enterprises, particularly SMEs, is likely to exert a "demonstration" effect on newly established businesses. Therefore, in the context of this study, should Chinese local governments opt to diminish various subsidies to companies as a response to fiscal pressures, such a course of action is anticipated to adversely impact the entry of new firms (Accetturo, 2022).

H3. Local fiscal pressure, by reducing subsidies for local SMEs,

negatively impacts the entry of new businesses.

In times of fiscal distress, governments may intensify tax collection efforts as a strategy to generate additional revenue (Peng et al., 2023; Yu & Mai, 2022). Yet, entrepreneurial activities are notably sensitive to taxation, and an increased tax burden will markedly affect the establishment of businesses (Belitski et al., 2016; Da Rin et al., 2011). Hence, it is reasonable to conjecture that under fiscal pressure, local governments may intensify tax collection efforts, thereby impeding the entry of new firms.

H4. Local fiscal pressure, through the enhancement of tax enforcement, negatively impacts the entry of new businesses.

Empirical studies have provided evidence that an optimized business environment, encompassing factors such as improved governmental governance (Kaufmann et al., 2009), enhanced legal system construction (Sobel, 2008), a cheap and fast incorporation process (Klapper et al., 2011) and a decrease in corruption (Kong & Qin, 2021), can foster entrepreneurial activities. Correspondingly, scholars have also highlighted that an adverse business environment can act as an impediment to the entry of new enterprises (Hallward-Driemeier, 2009; Klapper & Richmond, 2011; Muzi et al., 2023). Furthermore, amidst financial constraints, the government might resort to actions detrimental to the local business climate. In addition to previously discussed measures such as intensifying tax collection efforts, reducing subsidies, and creating disparities in financial access for enterprises, there may also be initiatives involving intervention in market pricing (Jiang et al., 2022) and a reduction in the provision of government services (Cooper, 1996; Scorsone & Plerhoples, 2010). Therefore, does fiscal pressure affect the entry of new enterprises by harming the local business environment? To test this mechanism channel, we propose hypothesis 5.

H5. Local fiscal pressure, by deteriorating the business environment, negatively impacts the entry of new businesses.

3. Research design

3.1. Data and variables

The data analyzed in this article encompasses both regional and corporate levels. At the regional level, the study incorporates data from 281 prefecture-level cities spanning the period from 2011 to 2021, yielding a total of 2630 valid observations. The economic and fiscal indicators for each city are obtained from their respective annual statistical yearbooks or bulletins and the China City Statistical Yearbook for each corresponding year. The patent application and registration data for prefecture-level cities are sourced from the website of the China National Intellectual Property Administration, while the business registration data for enterprises are derived from the National Enterprise Credit Information Publicity System, which publishes corporate registration information.³ At the corporate level, to include financial information from enterprises of various sizes, particularly SMEs in prefecture-level cities, we integrated two distinct databases: the National Taxation Survey Database and the financial data of listed companies from the CSMAR database. We then matched these data sets to their respective prefecture-level cities based on the location of each enterprise. In the robustness tests, we revised the criteria for determining the location of the enterprises to their actual places of operation. Constrained by the limitations in data availability, some databases do not cover the entire span of our study period. As a result, we endeavored to align the research intervals of various types of company samples with the primary regression analysis. Furthermore, we excluded company

samples from the financial sector and removed samples with outliers from our dataset, specifically those instances where total assets, primary business revenue, or the total output value of the enterprise were less than or equal to zero. Firms with excessive data missing are excluded from the dataset as well.

Regarding the measurement of new business entry at the prefecture-level city, we followed the approach of Kong and Qin (2021). This involved adding one to the number of newly registered enterprises per 10,000 people in each city annually, and then taking the natural logarithm of this figure to derive the business entry index (*Entry1*). In calculating the number of business entries for each city, we determined the corresponding city for each company based on the location of registration specified in each registration record. Additionally, in the robustness tests, we considered the actual operating locations of the companies to accurately match them with their respective cities. We also referenced Feng and Li (2023), directly using the natural logarithm of the number of business entries in each prefecture-level city as an alternative measure of business entry (*Entry2*).

In measuring the core explanatory variable of this study, the fiscal pressure of prefecture-level city governments, we adopt the mainstream “fiscal revenue-expenditure gap method” (Bai et al., 2019; Zhang et al., 2023; Zhu et al., 2023), which characterizes fiscal pressure using the difference between general public budget expenditures and general public budget revenues. We then normalized the fiscal gap using general public budget revenues to derive the fiscal pressure index (*FP1*), with reference to Liao and Liu (2014) and Zhu et al. (2020). For robustness, we also standardized it against the city’s GDP to calculate an alternative indicator (*FP2*), as suggested by Sun and Zhang (2019) and Gong et al. (2022). Furthermore, relying solely on the difference between fiscal expenditure and revenue to construct a fiscal stress indicator may be seen as overly simplistic, as this method can overlook critical factors such as local land revenue, government fund income, debt balances, and expenditure obligations from public-private partnership (PPP) projects, potentially leading to imprecise measurements. Therefore, following the methodology proposed by Ma and Qin (2022), we consider a third approach to construct the fiscal stress indicator (*FP3*), with the specific formula outlined as follows.

$$\text{Fiscal pressure (FP3)} = (\text{Debt}_{c,t} + \text{PPP}_{c,t}) / (\text{Rev}_{c,t} + \text{Fund}_{c,t}) \quad (1)$$

Where, $\text{Debt}_{c,t}$ refers to the outstanding debt balance of city c in year t . This data is obtained by aggregating the general-purpose and special-purpose local government debts starting from 2015. $\text{PPP}_{c,t}$ captures the annual payment obligations of the government in city c under PPP projects. $\text{Rev}_{c,t}$ represents the general public budget revenue of the prefecture-level city government, while $\text{Fund}_{c,t}$ denotes the government fund revenue. However, due to data availability constraints for the relevant indicators, the fiscal stress measure derived from this method covers a relatively short period, with valid data only available from 2015 to 2020. Therefore, this measure is used solely for robustness checks.

In order to mitigate estimation biases caused by omitted variables, a range of urban characteristics that could influence business entry are controlled for. GDPgth denotes the annual growth rate of a city’s GDP, calculated as the ratio of current period’s GDP to that of the previous period, and then subtracting 1. PCG is the natural logarithm of per capita GDP. Savings represents the logarithmic value of savings held by urban and rural residents. Popu is the natural logarithm of a city’s population. Area is the natural logarithm of a city’s land area. Second and Third respectively represent the proportions of the added value of GDP in the secondary and tertiary industries. Patent refers to the natural logarithm of the number of patents obtained by a city each year, representing the city’s level of innovation. FD represents the ratio of total deposits in financial institutions to GDP, serving as an approximate measure of a city’s level of financial development. Urban is the urbanization rate of a city, calculated by dividing the urban population by the total

³ The enterprises involved in the data are predominantly SMEs, which constitute the largest segment of business entities in China today (Lu et al., 2022).

population. *Salary* refers to the natural logarithm of the average salary of employees in a city. *House* is the annual average house price in urban areas. *Open* denotes the level of a city's openness, calculated as the amount of actual foreign capital used divided by GDP. *Trade* is the volume of a city's foreign trade, calculated by dividing the total import and export value by GDP. Moreover, to eliminate the influence of outliers, we conducted winsorization on all continuous variables at the 1 % and 99 % tails of their distribution. Table 1 presents the descriptive statistics for all variables in the main regression.

3.2. Empirical strategy

This study primarily employs a two-way fixed effects model for analysis, with the baseline regression model configured as follows.

$$\text{Entry}_{c,t} = \beta_0 + \beta_1 \text{FP}_{c,t} + \gamma \text{City_Controls}_{c,t} + u_c + \tau_t + \varepsilon_{c,t} \quad (2)$$

Where, *City_Controls* refers to the series of urban characteristic variables previously mentioned, *u* represents the city fixed effects, and τ denotes the year fixed effects. We clustered robust standard errors at the city level. Moreover, in the robustness tests, we also considered the scenarios where explanatory variables or control variables are lagged by one period.

Admittedly, the primary regression model is still subject to a certain degree of endogeneity, such as simultaneity, even though relevant confounding variables have been controlled for in the model. The most notable manifestation of this is that a deterioration in local economic conditions can, on one hand, exacerbate fiscal pressure on local governments, while on the other hand, it may also inhibit the entry of new enterprises. To mitigate these concerns of endogeneity, we designed a natural experiment to further validate the reliability of the results obtained in the main regression. Referencing Zhang and Lin (2022) and Qi et al. (2022), we adopt the VAT revenue sharing reform as a policy shock, which exerted considerable fiscal pressure on local governments in China. As a significant reform in the fiscal and taxation system, the transformation from Business Tax to VAT in China was initiated as a pilot program in 2012 and achieved full implementation by 2016. This reform, on one hand, reduced the issue of double taxation for enterprises, thereby alleviating their financial burden to some extent. On the other hand, it also resulted in a decrease in government fiscal revenues. However, the effect of reduced tax revenues was not pronounced before the full implementation of the Business Tax to VAT (B2V) reform. During the pilot phase, the reform was applied only to certain industries in specific regions, and the revenues that would have been collected as business tax continued to be allocated to local governments. On April 30, 2016, the State Council of China issued the "Transitional Plan for

Adjusting the Central and Local Division of Value-Added Tax Revenues after the Comprehensive Promotion of the Business Tax to VAT Pilot Program." This plan stipulated that the VAT paid by all enterprises would be included in the shared revenue pool between the central and local governments, with a fifty-fifty revenue sharing arrangement between the central and local authorities. The regulation was implemented concurrently with the comprehensive roll-out of the Business Tax to VAT reform, resulting in a sudden escalation of fiscal pressure on local governments. For local governments, the primary reasons for the substantial reduction in tax revenue are threefold: firstly, the tax reduction effect inherent in the Business Tax to VAT reform itself; secondly, a portion of the business tax revenue, which was originally entirely allocated to local governments, was also included in the shared tax revenue category between the central and local governments; thirdly, the central government claimed half of the remaining VAT revenue. Fig. 2 illustrates the changes in VAT and business tax revenues for local governments in China before and after the reform.

The shaded area in the diagram represents the proportion of revenue obtained by local governments from business tax and value-added tax.

The VAT revenue sharing reform was initiated by the Central Government of China as part of a fiscal and taxation system overhaul aimed at further alleviating the tax burden on enterprises. This reform was independent of the short-term economic growth conditions and the fiscal status of local governments at the time, and it was unrelated to the decision-making processes regarding business entry. Studies such as Zhang et al. (2021) and Liu et al. (2022) have confirmed that this reform indeed intensified the fiscal pressure on local governments in China, particularly at the prefecture-level city tier. Moreover, cities with a larger proportion of business tax revenue before the reform experienced a more pronounced fiscal pressure shock (Dong & Bai, 2017; Jiang & Yue, 2019). Consequently, on this basis, we can establish a quasi-natural experiment to assess the impact of fiscal pressure on the entry of new local enterprises, by considering the reform as an exogenous shock. Cities that were minimally affected by the policy are designated as the control group, whereas those that experienced more substantial impacts are identified as the treatment group. To facilitate appropriate grouping of the city samples, we first devised an index, Shock_c , to quantify the intensity of the policy shock, and then grouped the samples based on this index.

$$\text{Shock}_c = (\text{Business Tax}_{c,2015} + \text{VAT}_{c,2015}) / \text{GDP}_{c,2015} \quad (3)$$

Where, $\text{Business Tax}_{c,2015}$, $\text{VAT}_{c,2015}$,⁴ and $\text{GDP}_{c,2015}$ ⁵ represent the revenues from business tax and VAT, and the GDP of city *c* in the year 2015, respectively. The larger the value of this index, the higher the proportion of business tax in the city, implying a greater degree of impact from the policy shock. Moreover, cities with a Shock_c value in the top 30th percentile of its distribution are classified into the treatment group ($\text{Treat} = 1$); whereas those with a Shock_c value in the bottom 30th percentile are assigned to the control group ($\text{Treat} = 0$). The specific DID model corresponding to the natural experiment is formulated as follows.

⁴ VAT revenue is included in the calculation because several cities initiated the B2V reform before 2016, leading to the substitution of business tax with VAT in these areas. If the numerator of Shock_c only accounts for business taxes, this indicator would be underestimated. Furthermore, prior to the B2V reform, the contribution of VAT to local government tax revenues was approximately 15 %, while business tax, as a major source of local tax revenue, accounted for more than 30 %. Therefore, the Shock_c index primarily reflects the proportion of business tax within urban fiscal structures.

⁵ GDP is employed as the denominator for the indicator, rather than fiscal revenue. This is because, for cities that piloted the B2V reform prior to 2016, there has been a marginal decrease in fiscal revenue. Utilizing fiscal revenue as the denominator would, to some extent, overestimate the indicator.

Table 1
Summary statistics for the main variables.

Variables	N	S.D.	Mean	Min	Median	Max
Entry1	2630	0.892	4.112	2.195	4.170	6.253
Entry2	2630	1.112	10.029	6.557	10.009	13.650
FP1	2630	1.550	1.713	0.008	1.279	12.083
FP2	2630	0.085	0.112	-0.067	0.094	0.634
FP3	754	1.213	1.817	0.243	1.500	8.879
GDPgth	2630	0.081	0.087	-0.230	0.088	0.305
PCG	2630	0.670	10.733	9.333	10.665	12.501
Savings	2630	0.884	16.367	14.198	16.306	18.891
Popu	2630	0.646	5.938	3.584	5.964	7.217
Area	2630	0.793	9.326	7.375	9.377	12.112
Second	2630	0.102	0.466	0.159	0.471	0.722
Third	2630	0.099	0.417	0.208	0.412	0.721
Patent	2630	1.589	7.457	3.332	7.346	11.302
FD	2630	1.133	2.429	1.026	2.106	7.017
Urban	2630	14.268	56.492	26.754	54.637	94.976
Salary	2630	0.327	10.903	10.203	10.905	11.681
House	2630	0.468	8.675	7.918	8.577	10.285
Open	2630	0.016	0.017	0.0001	0.012	0.073
Trade	2630	0.264	0.179	0.001	0.083	1.475

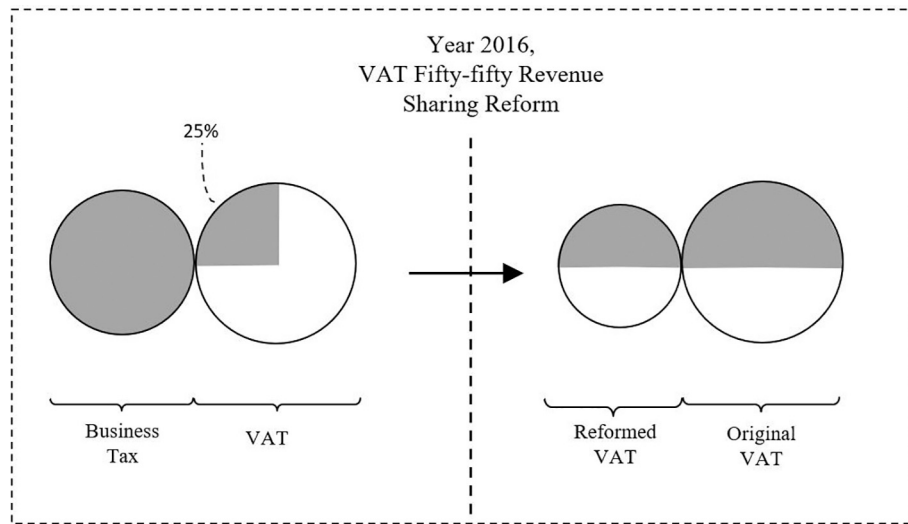


Fig. 2. Illustrative diagram of the decrease in tax revenue for local governments.

$$\text{Entry}_{c,t} = \beta_0 + \beta_1 \text{Treat}_c \times \text{Post}_{t,2016} + \gamma \text{City_Controls}_{c,t} + u_c + \tau_t + \varepsilon_{c,t} \quad (4)$$

In the model, $\text{Post}_{t,2016}$ denotes a binary variable that is set to 1 for the year 2016 and subsequent years, and 0 for prior years. The configuration of City_Controls , u_c , and τ_t is consistent with that in Eq. (2).

4. Empirical analysis

4.1. Baseline results

In this section, we commence with testing the core hypothesis of this paper, namely whether local fiscal pressure exerts a negative effect on the entry of new enterprises. Table 2 displays the regression results under various scenarios, where the core variables are measured using different methods. Columns (1) and (2) show the regression results with *Entry1* as the dependent variable, while Columns (3) and (4) to those with *Entry2*. Specifically, to more comprehensively reveal the relationships between the core variables, Columns (1) and (3) present the regression results without incorporating a series of city-level control variables, allowing for comparative analysis. The findings demonstrate that local fiscal pressure significantly inhibits new firm entry, as evidenced by the negative and statistically significant coefficients of the fiscal pressure variable at the 1 % level across all regression columns. Given the relative rationality and scientific rigor of the variable construction, we adopt the results using *Entry1* as the dependent variable and *FP1* as the independent variable as the baseline for subsequent analysis.

To strengthen the reliability of the baseline regression results, this study conducts several robustness tests. First, we re-estimate the regression using two alternative measures of fiscal pressure. From Table 3, we observe results consistent with those of the main regression. The regression coefficients of fiscal stress indicators *FP2* and *FP3* on the new firm entry variable are both negative and statistically significant at the 1 % and 5 % levels, respectively. Second, this study applies a lag structure to the relevant variables. On the one hand, the explanatory variables are lagged by one period. This approach not only helps to mitigate endogeneity concerns to some extent but also allows for a more in-depth exploration of whether fiscal stress exhibits any lagged effects. As shown in Column (1) of Table 4, although the statistical significance of the fiscal stress coefficient decreases slightly, it remains significantly negative, confirming the validity of the main findings. On the other hand, in light of potential disturbances from the previous period, all control variables are also lagged by one period. As indicated in Column

Table 2
Results of baseline regressions.

	Entry1		Entry2	
	(1)	(2)	(3)	(4)
FP1	−0.059*** (0.016)	−0.051*** (0.015)	−0.069*** (0.018)	−0.049*** (0.016)
GDPgth		0.129 (0.114)		0.129 (0.124)
PCG		0.207* (0.118)		0.182 (0.124)
Savings		−0.016 (0.102)		−0.187 (0.106)
Popu		−0.587** (0.290)		0.436 (0.299)
Area		−0.407 (0.259)		−0.511* (0.263)
Second		0.822 (0.566)		1.116* (0.603)
Third		0.939 (0.641)		1.252* (0.693)
Patent		0.012 (0.029)		0.014 (0.031)
FD		0.043* (0.024)		0.035 (0.026)
Urban		0.001 (0.004)		−4 × 10 ^{−4} (0.004)
Salary		0.378*** (0.133)		0.419*** (0.140)
House		0.009 (0.051)		0.017 (0.054)
Open		−3.334*** (0.897)		−3.252*** (0.920)
Trade		0.062 (0.096)		0.078 (0.102)
City Fixed Effects	YES	YES	YES	YES
Time Fixed Effects	YES	YES	YES	YES
Observations	2630	2630	2630	2630
Adj R ²	0.928	0.934	0.949	0.951

This table presents the regression results based on Eq. (2), with *FP1* as the dependent variable. Columns (1) and (2) use *Entry1* as the explanatory variable, while Columns (3) and (4) use *Entry2* as the explanatory variable. Additionally, Columns (1) and (3) present the regression results without including control variables, serving as a basis for comparison. Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

Table 3
Robustness results of the main regression - part A.

	Entry1	Entry2	Entry1	Entry2
	(1)	(2)	(3)	(4)
FP2	−0.957*** (0.317)	−0.903*** (0.330)		
FP3			−0.081** (0.035)	−0.081** (0.036)
GDPgth	0.171 (0.122)	0.171 (0.132)	−0.063 (0.159)	0.051 (0.162)
PCG	0.160 (0.122)	0.141 (0.128)	0.440* (0.245)	0.444* (0.248)
Savings	0.001 (0.101)	−0.003 (0.106)	−0.194 (0.130)	−0.188 (0.132)
Popu	−0.614** (0.294)	0.418 (0.304)	−1.225** (0.569)	−0.365 (0.598)
Area	−0.405 (0.259)	−0.512* (0.264)	1.272 (0.975)	1.259 (1.025)
Second	0.776 (0.558)	1.080* (0.595)	1.475** (0.704)	1.548** (0.753)
Third	0.875 (0.636)	1.196* (0.688)	1.347* (0.685)	1.387* (0.717)
Patent	0.019 (0.029)	0.021 (0.330)	0.171* (0.087)	0.178* (0.090)
FD	0.054** (0.025)	0.047* (0.026)	0.186** (0.077)	0.185** (0.078)
Urban	0.001 (0.004)	−2.00×10 ^{−4} (0.004)	−0.125 (0.008)	−0.012 (0.008)
Salary	0.389*** (0.131)	0.427*** (0.138)	0.317 (0.240)	0.315 (0.242)
House	0.022 (0.051)	0.030 (0.054)	−0.134 (0.105)	−0.117 (0.109)
Open	−3.125*** (0.879)	−3.324*** (0.902)	3.378*** (1.107)	3.291*** (1.105)
Trade	0.048 (0.096)	0.065 (0.102)	0.120 (0.114)	0.098 (0.125)
City Fixed Effects	YES	YES	YES	YES
Time Fixed Effects	YES	YES	YES	YES
Observations	2630	2630	465	465
Adj R ²	0.933	0.951	0.920	0.945

This table presents the results of robustness checks. To test the robustness of the findings reported in Table 2, alternative explanatory variables are introduced for further analysis. Following the regression framework in Table 2, the dependent variables remain *Entry1* and *Entry2*. Columns (1) and (2) use *FP2* as the explanatory variable, while Columns (3) and (4) replace it with *FP3*. Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

(2) of Table 4, the fiscal stress coefficient remains significantly negative, in line with the baseline regression results. Lastly, in identifying the cities where new firms have entered, we recalculated the entry indicator based on the city information provided in the firms' actual operating location rather than their place of registration. This adjustment is made because, in practice, some firms are registered in one location but operate in another. For such cases, using the actual place of operation to determine the city affiliation is more accurate. Accordingly, we employed text analysis techniques to extract city information from the companies' operating addresses. As illustrated in Columns (3) and (4) of Table 4, the results are consistent with the main regression, with the fiscal pressure coefficient remaining significantly negative. This further corroborates the suppressive effect of fiscal pressure on local new firm entry.

4.2. Alleviating issues of endogeneity

To address the potential issue of “simultaneity” endogeneity in the main regression model, where local economic conditions may simultaneously affect both local fiscal pressure and new enterprise entry as discussed in Section 3.2, we have devised a framework of natural experiment to further examine the robustness of the main regression's conclusions. Specifically, the VAT revenue sharing reform serves as the

Table 4
Robustness results of the main regression - part B.

	Entry1	Entry1	Entry2	Entry2
	Lagging the explanatory variable by one period	Lagging the control variables by one period	Defining a firm's affiliated city based on its operational location	
	(1)	(2)	(3)	(4)
FP1	−0.032* (0.018)	−0.052*** (0.016)	−0.048*** (0.015)	−0.049*** (0.016)
GDPgth	0.139 (0.130)	−0.087 (0.122)	0.143 (0.124)	0.119 (0.122)
PCG	0.192 (0.127)	0.173 (0.131)	0.175 (0.119)	0.156 (0.124)
Savings	−0.032 (0.108)	−0.033 (0.099)	0.052 (0.117)	0.083 (0.125)
Popu	−0.548* (0.307)	−0.281 (0.323)	−0.486* (0.290)	0.468 (0.298)
Area	−0.466* (0.276)	−0.433* (0.248)	−0.539 (0.267)	−0.560** (0.273)
Second	0.824 (0.643)	0.966* (0.560)	0.964* (0.569)	1.161* (0.595)
Third	0.852 (0.710)	1.061 (0.625)	1.007 (0.652)	1.215* (0.682)
Patent	0.022 (0.034)	−0.033 (0.032)	0.009 (0.032)	0.013 (0.032)
FD	0.040 (0.025)	0.043* (0.026)	0.037 (0.025)	0.037 (0.026)
Urban	0.002 (0.004)	0.005 (0.007)	0.001 (0.004)	−0.001 (0.004)
Salary	0.352** (0.151)	0.421*** (0.153)	0.417*** (0.137)	0.451*** (0.141)
House	−0.009 (0.055)	−0.006 (0.051)	−0.015 (0.059)	−0.028 (0.064)
Open	−3.241*** (0.985)	−5.683*** (1.046)	−3.041*** (1.018)	−3.093*** (1.015)
Trade	0.075 (0.107)	0.097 (0.113)	0.133 (0.106)	0.112 (0.106)
City Fixed Effects	YES	YES	YES	YES
Time Fixed Effects	YES	YES	YES	YES
Observations	2292	2292	2630	2630
Adj R ²	0.926	0.927	0.936	0.952

This table presents the results of an additional set of robustness checks. Column (1) reports the regression results when the explanatory variable *FP1* is lagged by one period. Column (2) reports the regression results when a series of control variables are lagged by one period. Columns (3) and (4) display the regression results using *Entry1* and *Entry2*, which are recalculated after redefining a firm's city affiliation based on its actual operating address, as the dependent variables. Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

policy shock in this natural experiment, as it represents a structural adjustment in China's fiscal and taxation system and is largely unrelated to short-term economic fluctuations across regions. To this end, as a preliminary test, we first need to verify whether a “parallel trend” exists between the control group and the treatment group. This requires that, prior to the policy shock, the two groups of cities exhibit no significant differences in new firm entry, after controlling for other city-level characteristics. Drawing on the event study methodology proposed by Jacobson et al. (1993), we design the following model to conduct the parallel trend test.

$$\text{Entry}_{c,t} = \beta_0 + \sum_{s=2011, s \neq 2015}^{10} \alpha_s \text{Treat}_c \times \text{year}_s + \gamma \text{City.Controls}_{c,t} + u_c + \tau_t + \varepsilon_{c,t} \quad (5)$$

In the equation, year_s denotes a dummy variable, assigned a value of 1 for the year s , and 0 for all other years. Here, 2015, the year preceding the implementation of the policy, is established as the base year. Appendix Table A.1 delineates the complete outcomes of the parallel

trends analysis. These results elucidate that prior to the reform, the control and treatment groups exhibit similar patterns regarding the entry of new enterprises. However, following the reform's implementation, a significant divergence emerged between the two groups, specifically manifested in a marked reduction in the scale of enterprises entering the treatment group. Fig. 3 visually presents the evolution of this trend. These findings substantiate the validity of the parallel trend assumption.

The values of the solid dots correspond to the coefficient α_5 in Eq. (5), and the vertical dashed lines represent the 95 % confidence intervals.

Table 5 reports the regression outcomes of the DID model, as specified in Eq. (4). Column (1) presents the outcomes based on the traditional DID model, where the coefficient of the $Treat \times Post$ term is significantly negative. To control for potential “self-selection” bias introduced by city-specific characteristics, we applied propensity score matching (PSM) to the sample cities and re-estimated the regression. The results, shown in Column (2), again indicate that the $Treat \times Post$ term remains significantly negative at the 5 % level. In addition, within the time frame covered in this study, some literature has mentioned certain events in specific years that may also impose fiscal pressure on local governments. To mitigate the influence of these time-specific events on the results of the DID model, we have accordingly shortened the study period to exclude the periods impacted by these events. First, some scholars have identified the 2010 “Educational Authority Reform” as a factor contributing to increased fiscal pressure on local governments (Ma & Yang, 2022; Xue et al., 2023). Accordingly, we limited the study period for the natural experiment to 2013–2021 and then re-conducted the DID model analysis. It can be observed that the coefficient of the $Treat \times Post$ term in column (3) of Table 5 continues to be negative and statistically significant. Second, in the studies by Green and Loualiche (2021) and Zhang et al. (2023), they argue that the COVID-19 public health event at the end of 2019 worsened the fiscal conditions of local governments in China, creating significant fiscal pressure. Moreover, the impact of the pandemic on small, medium, and micro-enterprises has introduced significant uncertainty to their business development, which could also hinder the establishment of new enterprises (She et al., 2020). To eliminate the impact of this event shock, we exclude post-2019 data from the analysis and re-run the regression. As shown in the results of Column (4), the conclusions remain unchanged.

To further assess the robustness of this natural experiment, we conducted both time placebo tests and individual placebo tests. In terms of time placebo test, we designated 2012, 2013, 2014 and 2015 as hypothetical policy implementation years and re-ran the regressions accordingly. The results, shown in Appendix Table A.2, indicate that the policy effects are insignificant across all hypothetical policy implementation years, thereby confirming that the results observed in Column

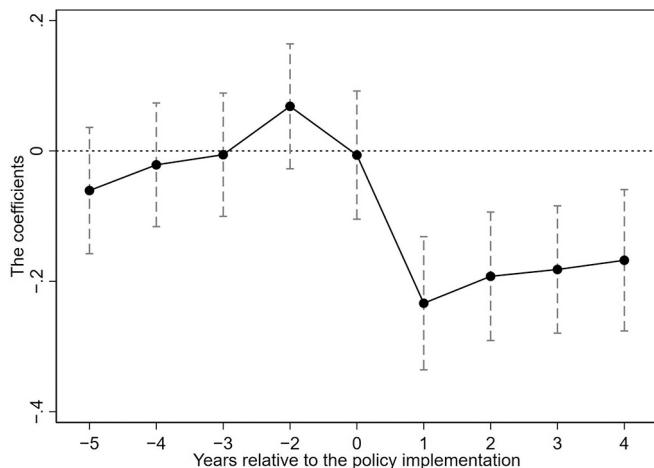


Fig. 3. Parallel trend test of the DID model.

Table 5
Results of DID model.

	Entry1			
	DID	PSM-DID	Excluding the impact of the “Educational Authority Reform”	Excluding the impact of the COVID-19 event
	(1)	(2)	(3)	(4)
Treat $\times$ Post	−0.114** (0.045)	−0.177** (0.089)	−0.115** (0.044)	−0.110** (0.428)
GDPgth	0.118 (0.141)	0.013 (0.147)	0.048 (0.147)	0.099 (0.167)
PCG	0.373** (0.153)	0.122 (0.202)	0.377** (0.171)	0.409** (0.169)
Savings	−0.065 (0.173)	0.010 (0.144)	−0.119 (0.197)	−0.130 (0.190)
Popu	0.059 (0.352)	−0.319 (0.455)	−0.011 (0.385)	0.234 (0.379)
Area	−0.491 (0.311)	−0.468 (0.348)	−0.572* (0.305)	−4.791 (0.305)
Second	0.140 (0.756)	0.032 (0.839)	0.331 (0.826)	0.182 (0.740)
Third	0.140 (0.819)	−0.883 (0.893)	0.220 (0.885)	0.362 (0.789)
Patent	0.012 (0.035)	0.075 (0.083)	0.062 (0.043)	−0.005 (0.030)
FD	0.037 (0.037)	0.062 (0.053)	0.035 (0.038)	0.049 (0.169)
Urban	3.207 $\times$ 10 ^{−4} (0.004)	1.351 $\times$ 10 ^{−4} (0.004)	−0.003 (0.004)	0.003 (0.005)
Salary	0.367** (0.169)	0.622** (0.257)	0.408** (0.212)	0.281* (0.167)
House	0.106 (0.074)	0.070 (0.099)	0.052 (0.078)	0.089 (0.062)
Open	−3.147*** (1.189)	−2.752* (1.454)	−1.194 (1.223)	−3.419*** (1.182)
Trade	0.022 (0.111)	0.390* (0.201)	0.064 (0.111)	0.073 (0.107)
City Fixed Effects	YES	YES	YES	YES
Time Fixed Effect	YES	YES	YES	YES
Observations	1743	1010	1375	1579
Adj R ²	0.934	0.899	0.919	0.941

This table reports the regression results based on the DID model specified in Eq. (4). Column (1) presents the original regression results. Column (2) shows the results after applying PSM to the variables. Column (3) provides the results after excluding data from periods affected by the “Education Governance Reform,” while Column (4) reports the results after removing data from periods influenced by the “COVID-19 event.” Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

(1) of Table 5 are not temporally random. In the individual placebo test, we arbitrarily assigned sample cities into treatment and control groups and conducted regression analysis analogous to that in Eq. (4). After repeating this process 500 times, we plotted the probability density functions of the coefficients for the $Treat \times Post$ interaction terms along with their p -values, as depicted in Fig. A.1 The illustration reveals that the coefficients follow a distribution resembling a normal curve centered around zero. Notably, the coefficient value of −0.114 lies in the tail region, suggesting that the results observed in Table 5 are randomly generated but are closely related to the assignment of the treatment group.

4.3. Mechanism analysis

4.3.1. Mechanism of credit crowding out

In this section, we will examine Hypothesis H2, which investigates whether, under the catalyst of fiscal pressure, local governments may

engage in biased resource allocation, encouraging financial institutions to extend more credit support to large enterprises. This preferential guidance tends to crowd out credit availability for local SMEs. As a result, newly established firms may face increased difficulty when applying for credit, thereby inhibiting new firm entry. To explore this issue, we assess the impact of fiscal pressure on credit accessibility for both large enterprises and SMEs at the local level. To ensure comprehensive coverage of local enterprise information and to enhance the robustness of our findings, we employ two distinct firm-level datasets: data from local publicly listed companies and data from enterprise tax surveys. However, due to the substantial differences in the amount of financial information provided by these enterprise databases, ensuring full consistency in firm-level control variables between the two datasets poses considerable challenges. We have endeavored to ensure consistency in firm-level control variables across the databases and rigorously account for factors that could potentially impact the dependent variable. Specifically, the firm-level control variables include: firm size (*Size*), revenue growth rate (*Growth*), return on assets (*ROA*), fixed asset ratio (*Fixed*), current asset ratio (*Current*), net commercial credit financing (*Commercial*), top ten shareholders' ownership ratio (*Top10*), and years since listing (*Age*). Furthermore, owing to the limited time coverage of the databases or the restricted availability of certain key data, studies based on these two databases may encounter inconsistencies in the research periods. To address this, we have made efforts to ensure that the research periods from both databases align with the time frame of the main regression analysis. Regarding city-level characteristics, we selected four key indicators for control in the regression analysis: annual GDP growth rate (*GDPgth*), per capita GDP (*PCG*), household savings levels (*Savings*), and the level of financial development in the city (*FD*).

Table 6 displays the regression results of fiscal pressure on corporate credit availability, based on data from the National Enterprise Tax Survey Database. Jointly conducted by China's State Administration of Taxation and the Ministry of Finance, this database is based on a sample survey of enterprises across the nation. Annually, it covers approximately 700,000 enterprises, including both large and small-to-medium-sized enterprises, encompassing various industries. After excluding samples from the financial sector and those with abnormal financial information, we obtained a dataset comprising 2,455,498 firm-year observations for the period 2011 to 2016. Furthermore, we classified firms based on the distribution of asset sizes in the dataset. Specifically, firms in the top 30 % in terms of total asset value were categorized as "large enterprises," those below the top 30 % were designated as "small and medium enterprises," and firms in the bottom 30 % were classified as "small enterprises." Considering the diverse industries represented in the tax survey database, categorizing firms by total asset size is a relatively straightforward and efficient approach to determining firm size. Additionally, constrained by the limited number of available indicators in the database, we constructed the enterprise bank credit availability index (*Bank loan*), using the formula (*liabilities – accounts payable*) / *total assets*, following the methodologies of Li et al. (2016) and Guo et al. (2023). The regression results in Table 6 indicate that the fiscal pressure coefficient for the large enterprise sample is significantly positive at the 10 % level. However, for the full sample, as well as the subsamples of "small and medium-sized only" and "small enterprises only," the coefficient is significantly negative at the 5 % level. This result suggests that under the influence of fiscal pressure, the gap in credit accessibility between large local enterprises and SMEs has further widened. Consequently, given the fixed availability of local credit resources, SMEs experience greater crowding out in their access to credit. This finding aligns with the theoretical proposition in Hypothesis 2, providing empirical support for it.

Table 7 presents the regression results of fiscal pressure on corporate credit availability, using data from publicly listed companies. In China, particularly at the prefecture-level city layer, publicly listed companies often serve as "core enterprises" or "major tax contributors" within the local industrial chain. Their scale typically far exceeds that of other local

Table 6
Results from tax survey database.

	Bank_loan			
	(Tax survey data:2011–2016)			
	Full Sample	Large size only	Small & medium size only	Small size only
	(1)	(2)	(3)	(4)
FP1	−0.130** (0.054)	0.009* (0.005)	−0.203** (0.085)	−0.609** (0.298)
Size	−0.796** (0.384)	0.034** (0.004)	−1.543** (0.701)	−4.420** (1.930)
Growth	8.29 × 10 ^{−8} (5.87 × 10 ^{−8})	−2.64 × 10 ^{−8} *** (3.48 × 10 ^{−9})	2.94 × 10 ^{−5} (3.23 × 10 ^{−5})	1.652 × 10 ^{−4} (3.143 × 10 ^{−4})
ROA	−0.003 (0.002)	−0.072** (0.029)	−0.003 (0.002)	−0.004 (0.002)
Fixed	0.207 (0.521)	0.093** (0.040)	0.850 (1.040)	5.996 (4.361)
Current	−1.792** (0.770)	0.067*** (0.024)	−1.945** (0.769)	−0.234 (0.826)
Commercial	−0.852*** (0.292)	−2.112*** (0.042)	−0.852*** (0.292)	−0.852*** (0.292)
GDPgth	−3.700* (2.036)	−0.018 (0.071)	−5.378* (3.190)	−19.890 (15.313)
PCG	−0.472 (0.806)	−0.004 (0.001)	−0.198 (1.162)	3.581 (6.410)
Savings	0.268 (0.683)	0.118** (0.053)	−0.393 (1.294)	−3.837 (7.151)
FD	−0.369* (0.204)	0.128 (0.012)	−0.558* (0.289)	−1.411* (0.712)
City Fixed Effects	YES	YES	YES	YES
Industry Fixed Effects	YES	YES	YES	YES
Time Fixed Effects	YES	YES	YES	YES
Observations	452,945	167,596	285,345	82,885
Adj R ²	0.378	0.006	0.378	0.380

This table presents the results of testing the "credit crowding-out" mechanism, based on tax survey data. Due to data availability constraints, the study period is limited to 2011–2016. As the database includes companies from various industries, there is no standardized criterion to classify firms by size. Therefore, following common practice, we categorize the sample firms by size based on their total assets. Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

enterprises, effectively classifying them as "large enterprises." Therefore, when categorizing the scale of sample enterprises in the dataset, we classify those in the lowest 10th percentile of total asset distribution as "small and medium enterprises," while the remaining enterprises are categorized as "large enterprises." In measuring credit availability, this study directly utilizes the logarithmic values of specific bank loan amounts disclosed by listed companies in their public announcements⁶ as the measurement indicator. Additionally, it is important to note that listed companies have a broad range of credit acquisition channels, which may include local banks as well as banks from the same province, other provinces, or even abroad. However, that the influence of prefectural city governments on financial institutions is confined to the local scope. Therefore, we select loans from local banks as the measure of "local" credit availability for the listed companies (*Local loan*). Here, "local banks" refer to those lending institutions that operate in the same locations as the registered offices of the listed companies. The results

⁶ Data from bank loan announcements of listed companies in the CSMAR database. Constrained by the availability of data, the research interval for the study here is set from 2013 to 2021.

Table 7
Results from listed company database.

	Local_loan		
	(Listed company data:2013–2021)		
	Full Sample	Large size only	Small & medium size only
	(1)	(2)	(3)
FP1	0.104 (0.070)	0.112* (0.065)	−1.570** (0.758)
Size	0.716*** (0.032)	0.707*** (0.031)	0.629** (0.294)
Growth	-9.88×10^{-5} (5.00×10^{-4})	-4.74×10^{-5} (5.072×10^{-4})	−0.035 (0.069)
ROA	0.402* (0.231)	0.565** (0.270)	−0.308 (0.624)
Fixed	0.592** (0.278)	0.669** (0.278)	1.338 (1.150)
Current	1.293*** (0.201)	1.402*** (0.206)	−0.416 (0.680)
Top10	0.204 (0.135)	0.173 (0.133)	0.497 (0.563)
Age	−0.016*** (0.004)	−0.016*** (0.004)	0.015 (0.016)
GDPgth	0.038 (0.304)	−0.001 (0.309)	0.802 (2.129)
PCG	0.355* (0.198)	0.400* (0.207)	−1.426 (1.202)
Savings	0.231** (0.090)	0.240** (0.094)	−3.013* (1.519)
FD	0.078* (0.044)	0.102** (0.048)	−0.218 (0.217)
City Fixed Effects	YES	YES	YES
Industry Fixed Effects	YES	YES	YES
Time Fixed Effects	YES	YES	YES
Observations	7900	7605	273
Adj R ²	0.326	0.304	0.380

This table presents the results of testing the “credit crowding-out” mechanism, using data from publicly listed companies. Due to data availability constraints, the research period is restricted to 2013–2021. Firm size is classified according to total assets. Given that publicly listed companies generally have larger asset scales, firms are divided into two categories: “large enterprises” and “small and medium-sized enterprises,” with “large enterprises” accounting for the majority of the sample. Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

from Table 7 reveal that within the sample of large enterprises, the coefficient of fiscal pressure is positive and significant, while in the sample of small and medium enterprises, it is significantly negative. This finding suggests that fiscal pressure exacerbates the disparity in credit availability between large enterprises and SMEs. It further confirms Hypothesis 2, that fiscal pressure on local governments leads to the crowding out of SMEs’ credit resources by large enterprises, thereby hindering new firm entry.

4.3.2. Mechanism of government subsidy

To test Hypothesis 3, we conducted regression analyses of the impact of fiscal pressure on the government subsidies received by enterprises. We develop two indicators regarding government subsidies to enhance the robustness of our conclusions. The first involves taking the natural logarithm of the subsidy amounts received by enterprises (*Subsidy1*). The second measure entails normalizing the amount of government subsidies by the enterprise’s operating revenue (*Subsidy2*). Moreover, since the majority of newly established local enterprises in our study are small or micro enterprises, and government subsidies to small enterprises may have a more pronounced “demonstration effect” on their entry, we focus our analysis on small enterprises using tax survey data and small and medium-sized enterprises from listed company datasets. The results from Table 8 indicate that across both datasets and regardless of the measurement method used, the coefficient of fiscal pressure is

Table 8
Regression results of government subsidy mechanism.

	Subsidy1	Subsidy2	Subsidy1	Subsidy2
	(Tax survey data: 2011–2016 Small size only)		(Listed company data: 2012–2015 Small & medium size only)	
	(1)	(2)	(3)	(4)
FP1	0.017 (0.022)	−0.002 (0.003)	0.194 (0.551)	0.0002 (0.019)
Size	0.043*** (0.006)	0.013*** (0.004)	1.414*** (0.310)	−0.008 (0.018)
Growth	-9.59×10^{-6} (2.57×10^{-6})	-3.06×10^{-7} (3.37×10^{-7})	-4.213×10^{-4} (0.004)	1.536×10^{-4} (1.451 $\times 10^{-4}$)
ROA	-9.03×10^{-6} (4.69×10^{-6})	6.06×10^{-6} (3.84×10^{-6})	0.425 (0.606)	0.013 (0.030)
Fixed	−0.093** (0.041)	−0.044 (0.075)	0.340 (0.870)	−0.044 (0.070)
Current	−0.007 (0.039)	−0.053 (0.066)	1.388 (0.855)	−0.045 (0.061)
Top10			1.117* (0.580)	−0.004 (0.038)
Age			−0.012 (0.025)	−0.001 (0.001)
GDPgth	0.462 (0.375)	0.045 (0.062)	−0.349 (3.603)	0.130 (0.175)
PCG	0.607** (0.262)	0.065** (0.030)	0.256 (0.955)	0.018 (0.077)
Savings	−0.542 (0.342)	−0.004 (0.039)	1.031 (1.890)	0.110 (0.079)
FD	0.047 (0.069)	0.002 (0.016)	0.043 (0.284)	−0.012 (0.021)
City Fixed Effects	YES	YES	YES	YES
Industry Fixed Effects	YES	YES	YES	YES
Time Fixed Effects	YES	YES	YES	YES
Observations	84,555	84,555	373	394
Adj R ²	0.130	0.003	0.320	0.162

This table presents the results aimed at examining whether fiscal pressure impacts the entry of new local enterprises by affecting government subsidies to small and medium-sized enterprises. To enhance the robustness of our conclusions, we conducted empirical analyses using firm-level samples from two different databases. Columns (1) and (2) present results based on the tax survey database, while Columns (3) and (4) report findings derived from the listed companies database. Constrained by the limited availability of data on government subsidies for enterprises, the study’s time frame was inevitably shortened. Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

not statistically significant. This suggests that Hypothesis 3 is not empirically supported.

4.3.3. Mechanism of tax enforcement

In Hypothesis 4, we propose that fiscal pressure may lead to an increase in local tax collection intensity, subsequently resulting in a higher tax burden on enterprises and impeding the entry of new businesses. To test this hypothesis, we first examine the relationship between fiscal pressure and local tax collection intensity. If a significant positive correlation is observed, we further investigate the relationship between tax enforcement and the entry of local enterprises. Referencing the approaches of Chen et al. (2016) and Chu et al. (2019), we develop the following two indicators to measure the tax collection intensity of prefectural cities.

$$TE1 = \frac{\text{Tax}_{c,t}}{\text{GDP}_{c,t}} / \frac{\widehat{\text{Tax}}_{c,t}}{\widehat{\text{GDP}}_{c,t}} \quad (6)$$

$$\frac{\text{Tax}_{c,t}}{\text{GDP}_{c,t}} = \beta_0 + \beta_1 \frac{\text{IND1}_{c,t}}{\text{GDP}_{c,t}} + \beta_2 \frac{\text{IND2}_{c,t}}{\text{GDP}_{c,t}} + \beta_3 \frac{\text{OPEN}_{c,t}}{\text{GDP}_{c,t}} + \varepsilon_{c,t} \quad (7)$$

Where, $\text{Tax}_{c,t}/\text{GDP}_{c,t}$, represents the ratio of the total tax revenue of city c at the end of year t to the local GDP. IND1 denotes the added value of the primary sector, and IND2 that of the tertiary sector. OPEN reflects the degree of openness of the region, indicated by the total import and export volume of each city for the respective year. $\text{Tax}_{c,t}/\text{GDP}_{c,t}$ is the fitted value obtained according to Eq. (7).

$$\text{TE2} = \frac{\text{Tax}_{c,t}}{\text{Tax}_{c,t}^*} \quad (8)$$

$$\text{Tax}_{c,t} = \beta_0 + \beta_1 \text{GDP}_{c,t} + \beta_2 \text{OPEN}_{c,t} + \beta_3 \text{density}_{c,t} + \beta_4 \text{IND2}_{c,t} + \beta_5 \text{IND3}_{c,t} + \beta_6 \text{urban}_{c,t} + \varepsilon_{c,t} \quad (9)$$

Where, *density* refers to the population density of the city, *IND3* denotes the added value of the tertiary sector, and *urban* represents the

Table 9
Test results of other mechanisms.

	TE1	TE2	Business1	Business2
	(1)	(2)	(3)	(4)
FP1	−0.082*** (0.017)	−0.075*** (0.014)	−0.004 (0.003)	0.015 (0.023)
GDPgth	0.080 (0.092)	0.429*** (0.121)	−0.112*** (0.018)	−0.308* (0.171)
PCG	−0.428*** (0.078)	0.451* (0.238)	0.064*** (0.016)	0.178 (0.159)
Savings	0.035 (0.061)	0.066 (0.063)	0.010 (0.013)	−0.164 (0.124)
Popu	−0.265 (0.219)	0.451* (0.238)	0.261*** (0.060)	−0.491 (0.410)
Area	−0.017 (0.441)	0.108 (0.205)	−0.148*** (0.049)	−0.121 (0.389)
Patent	−0.011 (0.018)	0.024 (0.020)	−0.011*** (0.004)	0.006 (0.051)
FD	0.019 (0.022)	0.024 (0.020)	0.006 (0.004)	0.023 (0.051)
Urban	−0.009 (0.006)	−1.625 × 10 ^{−4} (0.003)	0.001 (0.001)	−0.013** (0.006)
Salary	0.087 (0.113)	0.268*** (0.082)	0.013 (0.017)	−0.183 (0.228)
House	0.041 (0.043)	0.150** (0.063)	0.008 (0.007)	−0.028 (0.084)
Trade	−0.109* (0.063)	−0.094** (0.044)	−0.019 (0.019)	0.200 (0.153)
Second			0.223*** (0.077)	−1.457* (0.870)
Third			0.206*** (0.075)	−1.384 (0.876)
Open			0.012 (0.111)	1.788 (1.580)
City Fixed Effects	YES	YES	YES	YES
Industry Fixed Effects	YES	YES	YES	YES
Time Fixed Effects	YES	YES	YES	YES
Observations	2107	1290	2332	2556
Adj R ²	0.776	0.867	0.952	0.979

This table presents the results of tests examining alternative pathways through which fiscal pressure may impact the entry of new local firms. Columns (1) and (2) examine how fiscal pressure affects local governments' tax enforcement on enterprises, while Columns (3) and (4) analyze its impact on the local business environment index. Since the calculation of the tax enforcement intensity index already incorporates the variables for the proportion of value-added in the secondary industry (*Second*), the proportion of value-added in the tertiary industry (*Third*), and the level of openness to foreign trade (*Open*), to prevent multicollinearity between the control variables and the dependent variable in the regression analysis, we excluded these variables from the regression models in columns (1) and (2). Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

urbanization rate. Tax^* is the fitted value derived from Eq. (9). The results from Columns (1) and (2) of Table 9 show that the coefficient of fiscal pressure is negative and highly significant. This indicates that fiscal pressure, contrary to conventional understanding, actually weakens the intensity of local tax collection efforts. However, this also refutes the Hypothesis 4, as our expectation was that fiscal pressure would amplify local tax enforcement.

4.3.4. Mechanism of business environment

Lastly, regarding the hypothesis that fiscal pressure negatively impacts the local business environment, thereby inhibiting the entry of new enterprises, we conducted regression analyses as presented in Columns (3) and (4) of Table 9. In this context, *Business1* is an indicator used to measure the business environment, derived from the "Comprehensive Economic Competitiveness Index" in the "China Urban Competitiveness Report," as outlined by Yu and Liang (2019) and Zhou et al. (2020). *Business2* is a marketization index at the prefectural city level, constructed following the methodology of Fan et al. (2011). From the empirical results, it is observed that the coefficient of the explanatory variable *FP1* is not significant, thus refuting the assertion of Hypothesis 5.

5. Further exploration

In this chapter, building upon the primary conclusions, we extend our research to examine the heterogeneous impacts of company characteristics and regional features on the main findings, as well as the spatial spillover effects of fiscal pressure among cities on the entry of new enterprises.

5.1. Heterogeneity analysis

Firstly, we categorized the enterprise entry indicator based on the ownership type, distinguishing between state-owned and non-state-owned enterprises. At the prefectural city level, we calculated the annual entry counts for both types of enterprises separately. This classification yields two distinct indicators: state-owned enterprise entry (*Entry:Soe*) and non-state-owned enterprise entry (*Entry:Non-Soe*). Regression analyses were subsequently conducted to examine the impact of fiscal pressure on these two indicators and determine whether the state-owned characteristic introduces heterogeneous effects on the primary conclusions. As the literature suggests that state-owned enterprises possess an advantage in obtaining credit (Cull et al., 2009; Hau, Yi, Shan, & Sheng, 2019; Liu et al., 2021), this advantage may significantly alleviate the credit resource crowding-out effect typically experienced by SMEs, especially when the SMEs are also state-owned. In this case, the phenomenon of "entry obstruction" attributed to fiscal pressure is unlikely to occur or may be somewhat mitigated. The results from Columns (1) and (2) of Table 10 indicate that fiscal pressure does not have a significant linear relationship with the entry of state-owned enterprises, yet it exhibits a suppressive effect on the entry of non-state-owned enterprises. This aligns with our conjecture.

Secondly, we further elucidate the mechanistic role of governmental intervention in financial resources within its jurisdiction on the core findings of this study. To this end, we devised an indicator to quantify the degree of government intervention in the financial sector at the prefectural city level, which is then used to classify the sample cities. Initially, following the methodology of Liang and Wang (2021), we computed the scale of local government debt at the prefectural city level for each year from 2011 to 2014. This value was then normalized by the cities' annual GDP. Finally, we averaged these ratios over the four-year period to construct the government intervention (*GI*) indicator. Before 2015, the new Budget Law of China (revised in 2014) had not yet been implemented, so local governments largely lacked the authority to issue bonds independently. Most of their debt was accumulated through local government financing vehicles, and was predominantly absorbed by

Table 10
Results of heterogeneity analysis.

	Entry1					
	Entry:Soe	Entry:Non-Soe	Low GI group	High GI group	Low FI group	High FI group
	(1)	(2)	(3)	(4)	(5)	(6)
FP1	−0.002 (0.017)	−0.051*** (0.014)	0.001 (0.028)	−0.057*** (0.024)	−0.053*** (0.016)	0.039 (0.063)
Controls	YES	YES	YES	YES	YES	YES
City Fixed Effects	YES	YES	YES	YES	YES	YES
Time Fixed Effects	YES	YES	YES	YES	YES	YES
Observations	2630	2630	915	756	665	1024
Adj R ²	0.524	0.930	0.953	0.929	0.920	0.924

Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

local banks or financial institutions (Lu & Sun, 2013). As a result, local credit resources are occupied by local governments, either because financial institutions, perceiving government debt as a lower risk investment, choose to absorb it voluntarily, or because they are coerced into doing so under administrative pressure exerted by local authorities, as detailed by Huang et al. (2020). Thus, the *GI* indicator can, to some extent, reflect the capability of local governments to mobilize local financial resources. Accordingly, we categorized the sample cities based on this indicator. Cities that fall at or below the 30th percentile of the indicator's distribution are classified as the “low government intervention” group, while those at or above the 70th percentile are designated as the “high government intervention” group. The regression results in Columns (3) and (4) of Table 10 reveal that the regression coefficient of fiscal pressure on new enterprise entry is not significant in the low *GI* group, while it is significantly negative in the high *GI* group. This suggests that in areas with stronger governmental intervention in finance, the inhibitory effect of fiscal pressure on new enterprise entry is more pronounced, thereby reaffirming the mechanism proposed in Section 4.3.1.

Thirdly, we analyze the heterogeneous effects that the regional development level of inclusive finance brings to the primary conclusions of our research. Given the widespread penetration of the digital economy in China's prefecture-level cities (Zhang et al., 2024), the integration of digital technology with the financial sector is likely to enhance the inclusive provision of financial resources, potentially alleviating the financial accessibility constraints faced by local enterprises. Scholars have also confirmed that the development of inclusive finance within a region can mitigate the “credit discrimination” faced by SMEs (Bao & Huang, 2021; Hau et al., 2019; Lu, 2018). Consequently, the disparity in credit availability between local large enterprises and SMEs, caused by the government's biased allocation of financial resources under fiscal pressure, may be mitigated. This, in turn, could weaken the suppressive effect on the entry of new enterprises. To facilitate this test, we employed the “Peking University Digital Inclusive Finance Index” to measure the degree of inclusive finance development in each region and categorize the sample cities accordingly. Specifically, we first calculate the average level of inclusive finance development in each city over the research period, denoted as *FI*. Cities in the top 30th percentile of this indicator's distribution are classified as the “high financial inclusion” group, while those in the bottom 30th percentile are categorized as the “low financial inclusion” group. Based on the results presented in Columns (5) and (6) of Table 10, it is observed that in the Low *FI* group, the suppressive effect of fiscal pressure on the entry of new enterprises remains significant. In contrast, such an effect is not observed in the High *FI* group. The finding aligns with our expectations and confirms the mitigating impact of the level of inclusive finance development.

5.2. Spatial spillover effects

This study addresses the topic of new firm entry, a phenomenon that

exhibits a pronounced spatial mobility pattern. This implies that focusing solely on the local perspective, while neglecting potential impacts from other regions, would impose significant limitations on the analysis. Furthermore, it is necessary to examine whether there exists a spatial correlation between two core variables: fiscal pressure and new firm entry. If such a correlation indeed exists, conclusions derived from traditional panel regression analysis could be biased. In addition, if local fiscal pressure inhibits the entry of new firms, does fiscal pressure in neighboring cities also produce spillover effects on local firm entry? Based on these considerations, this section will extend the analysis from a spatial spillover perspective.

Prior to conducting spatial econometric analysis, it is necessary to first ascertain whether spatial correlation exists in both the explanatory and dependent variables of this study, and then select an appropriate spatial model. To achieve this, we initially conduct a Moran's *I* test. As shown in Appendix Table A.3, the Moran's *I* indices for both the fiscal pressure indicator, *FP1*, and the enterprise entry indicator, *Entry1*, are strongly significant in each year of the study period, rejecting the null hypothesis of “no spatial autocorrelation.” This highlights the need to apply a spatial econometric model to test the main conclusions of this study, with an initial suggestion to consider using a Spatial Durbin Model. Fig. 4 illustrates the spatial correlation of the core variables of this study, exemplified by the year 2020.

The solid fitted line in the figure indicates that the fiscal stress indicator and the new firm entry indicator are both correlated with their respective spatial lag terms. Additionally, the Moran's *I* statistic and its corresponding *p* – value reject the null hypothesis of “no spatial correlation.”

Furthermore, we employ the Lagrange Multiplier test to examine the necessity and appropriateness of incorporating spatial autoregressive terms and spatially correlated error terms into the spatial model. As indicated by the results in Table 11, the null hypotheses of “no spatial lag of the dependent variable” and “no spatial autocorrelation in the residuals” are both rejected. Therefore, further consideration of the GNSM may be necessary. To determine the suitability of this model, we utilize Likelihood Ratio (LR) tests to compare the applicability of various models, with the results presented in Table 12. The comparison results confirm the relative superiority of this model in the given context.

Incorporating the above test analyses, the spatial model configuration for this study is as follows.

$$\text{Entry}_{c,t} = \beta_0 + \rho W \text{Entry}_{c,t} + \beta_1 \text{Press}_{c,t} + \theta W \text{Press}_{c,t} + \gamma \text{City_Controls}_{c,t} + u_c + \tau_t + \varepsilon_{c,t} \quad (10)$$

$$\varepsilon_{c,t} = \lambda W \varepsilon_{c,t} + v \quad (11)$$

Where, *W* denotes the spatial weight matrix. To enhance the robustness of our conclusions, we employ three different matrices in the regression analysis: the adjacency matrix, inverse distance matrix, and economic distance matrix. Here, the elements of the inverse distance

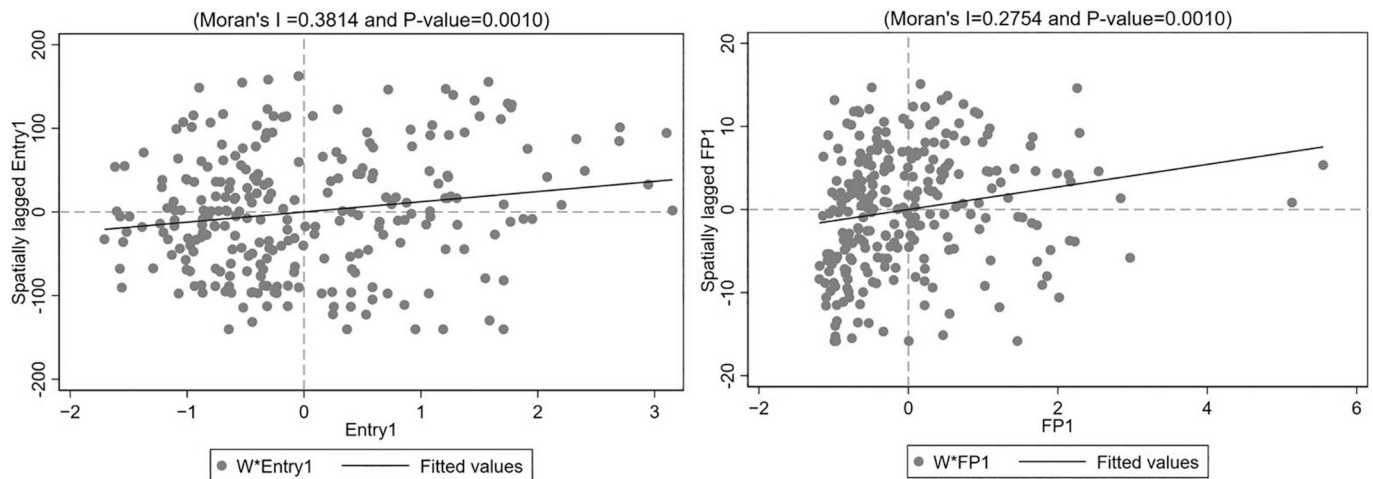


Fig. 4. Spatial correlation of core variables in 2020.

Table 11
Results of LM test.

	Spatial lag		Spatial error	
	Statistic	p-value	Statistic	p-value
LM test	5.950	0.015	106.636	0.000
Robust LM test	3.364	0.067	104.050	0.000

Table 12
Results of LR test.

GNSM vs SLX (Assumption: SLX nested within GNSM)	
LR $\chi^2 = 4236.69$	p-value = 0.0000
GNSM vs SDM (Assumption: SDM nested within GNSM)	
LR $\chi^2 = 3877.88$	p-value = 0.0000
GNSM vs SDEM (Assumption: SDEM nested within GNSM)	
LR $\chi^2 = 310.91$	p-value = 0.0000

Table 13
Results of spatial regression.

	Entry1		
	Using adjacency matrix	Using inverse distance matrix	Using economic distance matrix
	(1)	(2)	(3)
$W \times \text{Entry1}$	0.471*** (0.022)	0.411*** (0.143)	0.549*** (0.019)
FP1	-0.011* (0.006)	-0.036*** (0.005)	-0.019*** (0.005)
$W \times \text{FP1}$	0.032*** (0.009)	0.611*** (0.070)	0.056*** (0.007)
Controls	YES	YES	YES
City Fixed Effects	YES	YES	YES
Time Fixed Effects	YES	YES	YES
Observations	2340	2340	2340
Pseudo R ²	0.271	0.044	0.293

This table presents the regression results based on the spatial econometric models specified in Eq. (10) and Eq. (11). To strengthen the robustness of the findings, we perform multiple analyses using various adjacency spatial matrices. Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

matrix are calculated as the reciprocals of the spherical distances between locations, determined using their latitudinal and longitudinal coordinates. The elements of the economic distance matrix are defined as the reciprocals of the absolute differences in per capita GDP between two locations. The regression results are shown in Table 13.

It is observed that across all three columns, the coefficients of $W \times \text{Entry1}$ are consistently and significantly positive, confirming the spatial correlation associated with the entry of new enterprises. The negative significance of the coefficient for the FP1 term indicates that, even after accounting for the spatial lag of both the explanatory and dependent variables, the core conclusion of this study—that fiscal pressure inhibits the entry of new enterprises—remains valid. Notably, the coefficients of the $W \times \text{FP1}$ term are positively significant across the three result sets, suggesting that an escalation in fiscal pressures in neighboring cities promotes the entry of new enterprises within the focal city. Combining this with the inhibitory effect generated by local fiscal pressure, it reveals an “expulsion phenomenon.” Specifically, cities experiencing an increase in fiscal pressures tend to repel potential new enterprises that would have originally entered the locality, pushing them towards neighboring cities.

6. Conclusion

The entry of new businesses serves as one of the driving forces and a manifestation of vitality in regional economic growth. Local governments are keen to adopt various preferential or welfare policies to foster such entrepreneurial activities. However, in the face of fiscal pressures, certain actions by the government may become distorted, thereby exerting a countervailing force against the entry of new enterprises. This paper, employing a two-way fixed effects model for analysis, finds that local government fiscal pressure significantly impedes the entry of new enterprises in the locality. This finding has been corroborated through a series of robustness checks. To mitigate potential endogeneity concerns in regression analysis, we additionally design a natural experiment based on the policy shock from the VAT revenue sharing reform to further validate the reliability of our findings. Mechanistic analysis indicates that local governments, by intervening in the allocation of local financial resources, channel these resources preferentially towards larger enterprises in the area, consequently crowding out credit availability for SMEs. This, in turn, poses obstacles to the establishment of new businesses. We also exclude the potential channels through which fiscal pressure might influence the entry of new enterprises by affecting government subsidies, tax collection intensity, or the business environment.

Further analysis reveals that the state-owned nature of a business and

regional-level indicators such as the degree of government financial intervention and the level of inclusive financial development exert heterogeneous moderating effects on the main conclusion. Moreover, through spatial econometric analysis, we observe a spatial spillover effect of fiscal pressure on new enterprise entry. Specifically, when neighboring cities experience greater fiscal pressure, it tends to “propel” new enterprises to enter the focal city. Overall, this paper reveals a seemingly paradoxical phenomenon: while local governments aspire to encourage the entry of more new enterprises, their fiscal pressures lead them to mobilize local financial resources in a way that diminishes the financial accessibility for SMEs, thereby inadvertently impeding the establishment of new businesses. This reflects the unintended negative impact on local entrepreneurship by the government in its attempts to address fiscal difficulties. Our research endeavors to enrich the scholarly discourse on the determinants of enterprise entry and aims to offer a novel perspective in the study of the impact of local government fiscal pressure. Furthermore, this study contributes empirical evidence supporting the theory of the government’s “interventionist hand” and delves into analyzing its corresponding consequences. These findings offer valuable insights for the government to promote local entrepreneurship more effectively and sustainably. Admittedly, this study has certain limitations, primarily the lack of an economic mathematical model to support the arguments. Moreover, the empirical analysis faced considerable constraints due to data availability. These limitations

underscore areas that require further improvement in future research, particularly in exploring the micro-level impacts of fiscal pressure or the effects of government financial intervention.

Declaration of generative AI in scientific writing

During the preparation of this work the author employed ChatGPT to enhance the readability of the language used in this article. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Haiqiang Chen: Conceptualization, Methodology, Writing-review and editing, Supervision. Zhe Lin: Conceptualization, Data curation, Software, Writing-original draft, Writing-review and editing.

Acknowledgements

The authors would like to express their gratitude for the valuable comments provided by the anonymous reviewers. Haiqiang Chen appreciates the financial support from the National Natural Science Foundation of China with Grant Numbers #72233002 and #72173104.

Appendix A. Appendix

Table A.1
Results of parallel trend test of the DID model.

	Entry1
$Treat \times year_{-5}$	−0.061 (0.049)
$Treat \times year_{-4}$	−0.021 (0.048)
$Treat \times year_{-3}$	−0.006 (0.048)
$Treat \times year_{-2}$	0.068 (0.049)
$Treat \times year_0$	−0.006 (0.050)
$Treat \times year_1$	−0.234*** (0.052)
$Treat \times year_2$	−0.192*** (0.050)
$Treat \times year_3$	−0.182*** (0.050)
$Treat \times year_4$	−0.168*** (0.055)
$Treat \times year_5$	−0.545*** (0.096)
Controls	YES
City Fixed Effect	YES
Time Fixed Effect	YES
Observation	1743
Adj R ²	0.935

Robust standard errors clustered at city level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively. Controls here refer to all city-specific control variables listed in Table 5.

Table A.2

Results of time placebo test of the DID model.

	Entry1			
	(1)	(2)	(3)	(4)
<i>Treat</i> × <i>post</i> ..2012	−0.027 (0.032)			
<i>Treat</i> × <i>post</i> ..2013		−0.014 (0.033)		
<i>Treat</i> × <i>post</i> ..2014			−0.019 (0.036)	
<i>Treat</i> × <i>post</i> ..2015				−0.065 (0.085)
Controls	YES	YES	YES	YES
City Fixed Effect	YES	YES	YES	YES
Time Fixed Effect	YES	YES	YES	YES
Observation	1743	1743	1743	1743
Adj R ²	0.933	0.933	0.933	0.933

Robust standard errors clustered by city are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively. *Controls* here refer to all city-specific control variables listed in Table 5.

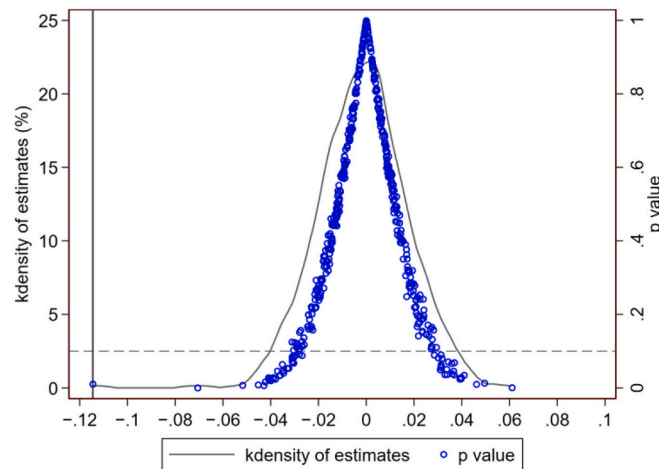


Fig. A.1. Placebo test for city samples. In the figure, after randomly assigning labels to the treatment and control groups 500 times, the resulting coefficients and their *p*-values exhibit a curve distribution around zero, resembling a normal distribution. Additionally, the estimated coefficient of −0.114 in Table 5 falls at the tail of the distribution, indicating that observing this coefficient value in a single instance is a low-probability event. These findings suggest that the results obtained from the DID model in the previous section are not randomly generated.

Table A.3

Results of Moran's I test.

	Entry1		FP1	
	Moran's I	z-value	Moran's I	z-value
2011	0.274***	13.057	0.191***	9.712
2012	0.251***	12.005	0.224***	10.976
2013	0.226***	10.835	0.259***	12.551
2014	0.231***	11.046	0.243***	11.789
2015	0.223***	10.689	0.240***	11.539
2016	0.232***	11.139	0.269***	12.861
2017	0.214***	10.239	0.263***	12.608
2018	0.215***	10.291	0.268***	12.833
2019	0.183***	8.767	0.263***	12.642
2020	0.381***	18.100	0.275***	13.255
2021	0.253***	12.096	0.245***	11.819

***, **, and * indicate significance at the 1 %, 5 % and 10 % levels, respectively.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Glossary

- Local fiscal pressure:** the financial strain experienced by local governments, specifically the imbalance between expenditures and revenues within a certain period due to liquidity challenges.
- New Firm Entry:** the registration and commencement of operations by newly established firms in a particular region. It is essentially reflected in the growth of newly registered firms, serving as an indicator of entrepreneurial vitality and regional dynamism.
- Governmental intervention:** the deliberate interference by the government in the allocation and management of economic resources within its jurisdiction.
- The reform of replacing business tax with value-added tax (B2V reform):** a significant transformation in China's fiscal and taxation system. This reform aims primarily to reduce the double taxation imposed on enterprises by substituting the business tax with a value-added tax.
- Tax enforcement:** defined in relevant literature and referenced in this paper, refers to the intensity with which the government executes taxation on taxable entities within its jurisdiction.
- Local government financing vehicles:** companies established by local governments in China with the primary purpose of raising funds. Prior to the 2014 revision of the Budget Law, most local governments in China were not legally authorized to borrow directly.